PROJECT DOCUMENT · TECHNOLOGY · PITCH

GestureLab CAD

A 3D engineering tutor you can talk to,
point at, and learn from, in your own language.

Voice: Gnani Prisma (speech-to-text) and Gnani Timbre (text-to-speech)

3D: Three.js · Hands: MediaPipe Hand Landmarker · Teacher: Gemini

Great Indian AI Internship Challenge 2026 by Gnani AI

Jugaad Genius Award entry: “the most creative or unexpected use of Gnani AI's models”

#GreatIndianAIInternshipChallenge #GnaniAI

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SECTION 01

The Pitch

THE ONE-LINE PITCH

GestureLab CAD lets every student point at a machine, ask it a question out loud in their own language, and watch the machine answer: the right parts light up and move while a natural voice explains.

The 60-second pitch

Every engineering student has looked at a machine (a gearbox, an engine, the steel inside a concrete column) and wondered: *why is this part here, and what does it do?* Textbooks give flat diagrams. 3D viewers show shape but explain nothing. Chatbots write paragraphs but cannot see the part the student means. And for millions of students who think in Telugu or Hindi, almost every answer arrives in English.

GestureLab CAD makes the machine itself the teacher. The student points at a part with a finger through an ordinary webcam, presses the microphone once and asks naturally. **Gnani Prisma** turns the speech into text and detects the language by itself, with no menu. An AI teacher plans a short lesson from the model's *real* parts data, every step is checked before it runs, and **Gnani Timbre** speaks each sentence at the exact moment its parts light up, in the same language the student used. Then the tutor checks understanding with a short spoken quiz.

It runs in a web browser, on a laptop, a webcam and a microphone. No lab, no expensive cut-away models, no installation.

Why it is more than a chatbot

A chatbot or video tutor…	GestureLab CAD…
answers in text beside a picture	**shows the answer on the machine** while it speaks
cannot see what you mean	**knows the part you point at** (“this gear“)
answers from general knowledge	**answers from the model's real parts data**; every action is validated
needs you to choose a language	**detects English, Telugu or Hindi automatically** and replies in it
only talks	**checks that you understood** with a short spoken quiz
is one more screen to read	is controlled by **voice and hand gestures together**

The creative use of Gnani AI (the “jugaad”)

In most apps, speech-to-text is a keyboard and text-to-speech is a loudspeaker. In GestureLab CAD, Gnani's voice models are the **controls of a 3D lesson**: Prisma turns a spoken question into a validated action on a real 3D machine, and Timbre is synchronised with motion, so hearing and seeing happen together. A laptop, a webcam and Gnani's Indian-language voice models stand in for an expensive cut-away machine and a tutor who speaks the student's language.

SECTION 02

The Problem

Students can **see** a machine, but they cannot **ask** it anything. The tools used to teach machines today each solve part of the problem, and none of them answers the student's own question at the moment it appears.

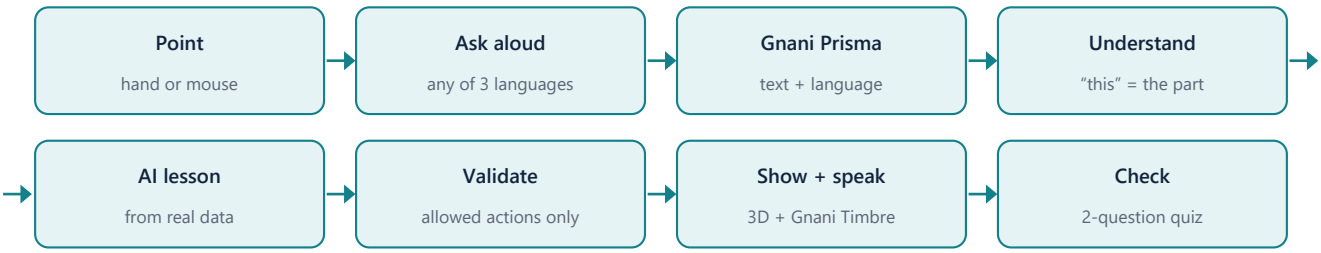
Pain point	What happens to the student
Flat textbooks and slides	A 2D cross-section cannot show how gears mesh or how a structure carries load, so students memorise diagrams instead of understanding mechanisms.
3D viewers explain nothing	Rotating a CAD model shows shape, not purpose. Professional CAD tools are costly and built for designers, not learners.
Cut-away models are rare	Physical models are expensive and fragile. In many colleges, polytechnics and ITIs, students never take a real machine apart.
No one to ask	Doubts appear late at night or during revision, when no teacher is available.
The language barrier	Many students think in Telugu or Hindi, but explanations, videos and tools are almost always in English.
Chatbots don't fix it	A text chatbot cannot see the part the student is looking at, cannot show anything on the machine, and does not know the parts of the model in front of them.

SECTION 03

What We Built

GestureLab CAD combines three things that normally live in separate tools (a 3D model, a voice conversation and a teacher's explanation) into one experience in the browser.

One question, end to end



#	Step	What happens
1	The learner chooses a part.	They point at it with a finger through the webcam, click it, or name it. The part lights up and the inspector shows its role, specifications, materials and what it drives, meshes with or supports.
2	The learner asks out loud.	One press of the microphone. The app detects when speech starts, allows natural pauses, and stops recording by itself when the learner finishes. Short replies like “Yes” are kept.
3	Gnani Prisma listens and detects the language.	Prisma v3.0 (automatic Indic language detection) returns the exact words and the language. The transcript appears as “You asked · Telugu: …” in the learner's own script, never translated. The model's part names are sent as key terms, so words like “sun gear” are recognised.

#	Step	What happens
4	The app understands “this”.	A deterministic parser resolves the question against the pointed, selected, named or last-discussed part. Simple requests (“zoom in”, “what is this?”) are answered instantly from the model data, with no AI call.
5	An AI teacher plans a grounded lesson.	For explanations, Gemini receives a fact pack (the model's real parts, roles, relationships and the current scene) and streams back a short lesson plan: each step pairs one or two spoken sentences with a 3D action.
6	Every step is validated.	Actions must come from an allowed list and name parts that really exist. They are checked on the server and again in the browser. The AI returns data, never code.
7	Gnani Timbre speaks while the machine moves.	Each sentence is spoken as its parts light up, focus or separate. The first sentence plays while the rest is still being written, so the answer starts within seconds.
8	The tutor checks understanding.	It asks whether the explanation made sense, offers to explain again, and can ask two short quiz questions, evaluating the learner's spoken answers.
9	The conversation stays natural.	Pressing the mic interrupts at once and no stale answer plays. Every question is detected in its own language, so a learner can switch English → Telugu → Hindi between turns.

Key features

Area	Features
Interactive 3D learning	Five learning modules: an epicyclic planetary gearbox and a steam engine (mechanical), a reinforced-concrete building frame and a Warren truss bridge (civil), and the human eye (biology). Exploded view and one-click reassembly; a colour-coded, see-through X-ray view with labels; an engineering inspector (role, specifications, materials, relationship map); an assembly tree to browse, search, select and hide parts.
Multilingual voice tutor	One press to talk; automatic stop; natural pauses and short replies handled; automatic English / Telugu / Hindi detection; answers spoken in the same language with technical terms kept in English; explanation synchronised with the 3D model; “explain again”; a two-question spoken quiz; interrupt any time.
Hand-gesture control	Open palm to orbit, pinch to zoom, point and hold to select, two hands to explode and reassemble. Voice and gesture work together: point at a part and ask about “this”.
Honest and safe by design	Each step is labelled *shown in the 3D model*, *from the model data*, *engineering explanation* or *not simulated in this app*. Forces and failures are explained as reasoning, never presented as simulation. If the AI teacher is unavailable, a grounded explanation is built from the model data instead.
Clear errors	A muted microphone, a denied permission, a busy device and a service error each produce their own clear message, never a generic “say it again”.

SECTION 04

Technology Stack at a Glance

Every technology in GestureLab CAD has one clear job. The browser handles 3D, hands and audio; the backend holds every key and talks to the AI services.

Layer	Technology	Purpose in one line
Frontend	React 19	Builds the interface out of reusable components and keeps it in sync with the app's state
	TypeScript 6	Adds types to JavaScript, catching mistakes before the app runs
	Vite 8	Fast development server and optimised production build
	Three.js	Draws and animates the interactive 3D models in the browser (WebGL)
	Web Audio API	Records the microphone, detects speech and encodes WAV audio in the browser
Hand tracking	MediaPipe Tasks Vision: Hand Landmarker	Pretrained models that find hands and 21 3D points per hand, in real time, in the browser
Backend	Python 3.12 + FastAPI	The voice and teaching API; keeps every API key on the server
	Uvicorn	The server that runs the FastAPI application
	Requests, python-multipart	Calls Gnani and Gemini; receives audio uploads
Speech AI	Gnani Prisma (stt/v3, Prisma v3.0)	Speech-to-text with automatic English / Telugu / Hindi detection
	Gnani Timbre (timbre-v2.5)	Natural speech in the learner's language, with a native voice per language
Reasoning AI	Gemini (gemini-3.5-flash)	Writes short lesson plans grounded in the model's data; checks quiz answers
Quality	Node test runner, pytest, oxlint	983 frontend and 222 backend automated tests, plus linting and type checks
Hosting	Vercel + Render	Frontend on Vercel, backend on Render, both over HTTPS (needed for mic and camera)

SECTION 05

Technology Stack in Detail

For each technology: what it is, why we chose it, and the exact job it does in GestureLab CAD.

Frontend (runs in the learner's browser)

React 19
UI LIBRARY

What it is · A JavaScript library for building user interfaces from small, reusable components.
Why we chose it · The app has many live, interacting panels (3D view, parts tree, inspector, voice dock, tutor card, camera view). React re-draws exactly what changed when the state changes, so the interface stays correct and responsive.
Its job in GestureLab · Renders every screen and panel: the workspace, the voice dock with its live states (listening, understanding, speaking), the "You asked" transcript, the tutor and quiz cards, and the gesture camera view.

TypeScript 6
LANGUAGE

What it is · JavaScript with static types, checked by a compiler before the code runs.
Why we chose it · The project has many data shapes that must match exactly: 3D model definitions, scene commands, lesson plans, API replies. Types catch mismatches at build time instead of in front of a student.
Its job in GestureLab · All frontend code is TypeScript. Scene commands, gesture types, lesson-plan steps and voice results are typed, so an invalid command cannot even be written.

Vite 8
BUILD TOOL

What it is · A modern development server and bundler for web apps.
Why we chose it · Starts instantly, reloads changes in milliseconds during development, and produces a small, optimised build for deployment. In development it also forwards /api calls to the local backend.
Its job in GestureLab · Runs the app locally, bundles it for production (the files deployed to Vercel), and serves the MediaPipe model and WebAssembly files.

Three.js
3D GRAPHICS

What it is · The most widely used JavaScript library for 3D graphics on the web, built on WebGL (the GPU in the browser).
Why we chose it · It renders detailed, animated 3D models smoothly in any modern browser, with no plug-in or installation, and gives full control over materials, cameras and animation.
Its job in GestureLab · Draws the five learning modules; animates the exploded view and reassembly; highlights, focuses and isolates parts; provides the X-ray view and labels; moves the camera for every lesson step.

Web Audio API
BROWSER AUDIO

What it is · The browser's built-in audio-processing system.
Why we chose it · Needed to capture the microphone with echo cancellation and noise suppression, and to analyse the sound in real time without sending anything until the learner has finished speaking.
Its job in GestureLab · Records the microphone; runs our speech detector (when speech starts, natural pauses, when it ends); detects a muted microphone; converts the recording to a 16 kHz mono WAV for Prisma; plays Timbre's answers in order.

Backend (runs on the server)

Python 3.12 + FastAPI
API FRAMEWORK

What it is · Python is the main language of AI and data work; FastAPI is a modern, fast Python framework for building web APIs.
Why we chose it · FastAPI is simple, fast and validates every request automatically. Most importantly, a backend keeps the API keys secret: the browser never talks to Gnani or Gemini directly.
Its job in GestureLab · Provides the app's API: /api/voice/transcribe (to Prisma), /api/voice/speak (to Timbre), the streamed teaching endpoint (to Gemini), the quiz-answer check and a health check. It also re-validates every lesson plan.

Uvicorn
WEB SERVER

What it is · A lightweight, high-performance server for Python web applications.
Why we chose it · It is the standard way to run FastAPI, and it supports streaming responses, which lets the first sentence of an answer reach the student before the rest is written.
Its job in GestureLab · Runs the backend locally and on Render.

Requests + python-multipart
LIBRARIES

What it is · Requests is Python's standard library for calling web services; python-multipart lets FastAPI receive file uploads.
Why we chose it · Simple, reliable and well tested.
Its job in GestureLab · Requests sends audio to Prisma, text to Timbre and prompts to Gemini (including streamed answers). python-multipart receives the learner's audio recording from the browser.

Artificial intelligence

Gnani Prisma
SPEECH-TO-TEXT

What it is · Gnani AI's speech recognition model for Indian languages (API stt/v3, model Prisma v3.0).
Why we chose it · It is built for Indian speech and languages, and it can detect the language automatically (“indic-auto”), including code-mixed speech. That is what lets a student simply speak, in English, Telugu or Hindi, with no menu.
Its job in GestureLab · Turns every spoken question into text and tells the app which language was spoken. The model's part names are sent as key terms, so technical words are recognised correctly.

Gnani Timbre
TEXT-TO-SPEECH

What it is · Gnani AI's speech synthesis model (timbre-v2.5), with natural native voices for Indian languages.
Why we chose it · Students understand best when they *hear* an explanation in their own language, from a natural voice. Timbre speaks Telugu and Hindi naturally while keeping English technical terms.
Its job in GestureLab · Speaks every reply and every lesson step in the learner's language, sentence by sentence, in sync with the 3D model.

Gemini
REASONING (LLM)

What it is · Google's large language model (gemini-3.5-flash), used through an OpenAI-compatible API.
Why we chose it · It can write a short, structured lesson plan in JSON, in the learner's language, quickly, and it streams its answer so speech can start early. We ask it for a low reasoning level, which in our live tests cut the time to its first words from about 14 s to about 3-4 s.
Its job in GestureLab · Writes lesson plans grounded in the fact pack (real parts, roles, relationships, scene); checks quiz answers. It returns data only: every action it suggests is validated before it runs.

Quality and hosting

Node test runner, pytest, oxlint QUALITY TOOLS	What it is · Automated test frameworks for TypeScript and Python, and a fast linter. Why we chose it · A voice tutor has many moving parts. Automated tests prove the behaviour after every change, without spending API credits (Gnani and Gemini are simulated in the tests). Its job in GestureLab · 983 frontend tests and 222 backend tests cover speech detection, microphone failures, language decisions, lesson validation, playback order, gestures and API error handling.
Vercel + Render HOSTING	What it is · Cloud platforms: Vercel serves websites; Render runs backend services. Why we chose it · Both deploy straight from GitHub and provide HTTPS, which browsers require before they allow microphone and camera access. Its job in GestureLab · Vercel serves the frontend and forwards every /api request to the Render backend, so the browser sees one site and the API keys stay on Render.

SECTION 06

3D Hand Gesture Recognition

The learner controls the 3D model with their bare hands, through an ordinary laptop webcam.

A CLEAR ANSWER: WHICH OPENCV PRETRAINED MODELS DO WE USE?
None. GestureLab CAD does not use OpenCV or any OpenCV model. Hand tracking uses **Google MediaPipe's pretrained Hand Landmarker**, running directly in the browser. On top of it, the gestures themselves are recognised by **our own rules** (geometry, smoothing and a state machine); no custom model was trained.

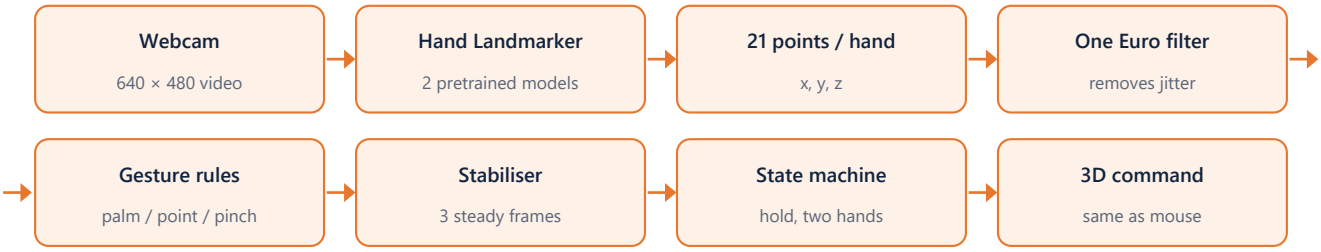
The pretrained models we use

We use MediaPipe's **Hand Landmarker** model bundle (`hand_landmarker.task`, float16). The bundle contains two pretrained neural networks that work as a pair:

Pretrained model	What it does	Why it matters here
1. Palm detection model	Scans the whole camera image and finds where hands are (a palm is easier to detect reliably than a whole hand with moving fingers).	Finds the learner's hand anywhere in view, and both hands for the two-hand explode gesture.
2. Hand landmark model	Looks only at the detected hand region and predicts **21 3D landmarks**: the wrist and four points on each finger, plus whether it is a left or right hand.	These 21 points are precise enough to measure a pinch, see which fingers are extended, and follow a pointing fingertip.

Once a hand is found, the landmark model follows it from frame to frame, and the palm detector runs again only when the hand is lost. While no hand is visible, we search about 15 times a second instead of on every frame, which halves the work the laptop does when idle.

How a hand movement becomes a 3D action



Stage	How it works
Camera	The browser opens the webcam (video only, never audio). The video stays on the device.
Hand Landmarker	MediaPipe runs in the browser through WebAssembly, on the GPU (WebGL) when available. Settings: video mode, up to 2 hands, confidence 0.5. The model files are served with the app, with Google's CDN as a fallback.
Smoothing	A One Euro filter (Casiez et al., CHI 2012) removes small jitters when the hand is still, but stays responsive when it moves fast.
Gesture rules	Our own geometry: which fingers are extended (tip-to-knuckle ratios), and the thumb-to-index distance divided by the hand's size, so a pinch works at any distance from the camera. Result: open palm, pointing, pinch or none.
Stabiliser	A gesture must be steady for 3 frames before it counts, and a pinch is released only when the fingers open wider than where it started (hysteresis), so gestures never flicker.
State machine	Turns gestures into actions: hold a point for 1.2 s to select; pinch both hands, then open them to explode; pinch both again to reassemble. While two hands are visible, one-hand gestures pause.
Command	Every gesture becomes the same validated scene command as a mouse click, keyboard shortcut or voice request.

The gestures

Gesture	Action
Open palm, move the hand	Orbit (rotate) the 3D model
Pinch thumb and index, move closer or apart	Zoom in and out
Point with the index finger, hold 1.2 s	Select the part under the fingertip
Pinch both hands, then open both hands	Explode the assembly into its parts
Pinch both hands again	Reassemble the model
Point at a part, then ask by voice	"What is *this* connected to?" refers to the pointed part

Why MediaPipe instead of OpenCV?

Requirement	MediaPipe Hand Landmarker	An OpenCV-based approach
Where it runs	Inside the browser (WebAssembly + GPU). Nothing to install.	Usually a Python program or server that must receive the webcam stream.
Privacy	Camera video never leaves the learner's device.	Video would need to be sent to a server for a web app.

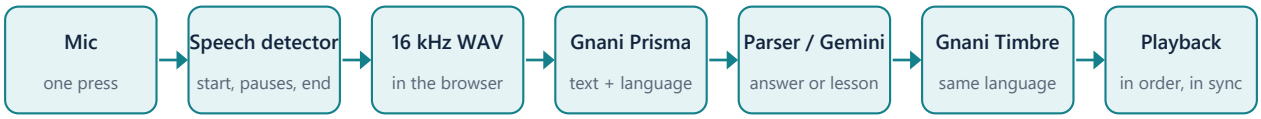
Requirement	MediaPipe Hand Landmarker	An OpenCV-based approach
Hand understanding	Pretrained, ready to use: 21 3D landmarks and left/right hand, built for real-time tracking.	OpenCV is a general image-processing toolkit; precise finger landmarks need a separate hand model added on top.
Speed on a normal laptop	Designed for real-time tracking on ordinary devices; tracks between frames instead of re-detecting.	Depends on the added model; streaming video to a server adds network delay.
Training data	None needed: the models are pretrained by Google on large hand datasets.	A custom hand model would need data and training.

In short: MediaPipe gave us accurate, private, real-time hand tracking in the browser with no training and no server, which is exactly what a low-cost classroom tool needs. Our own contribution is the layer on top: smoothing, gesture rules, the stabiliser and the state machine that turn 21 points into reliable 3D actions.

SECTION 07

The Voice Pipeline

From the moment the learner presses the microphone to the moment they hear the answer.

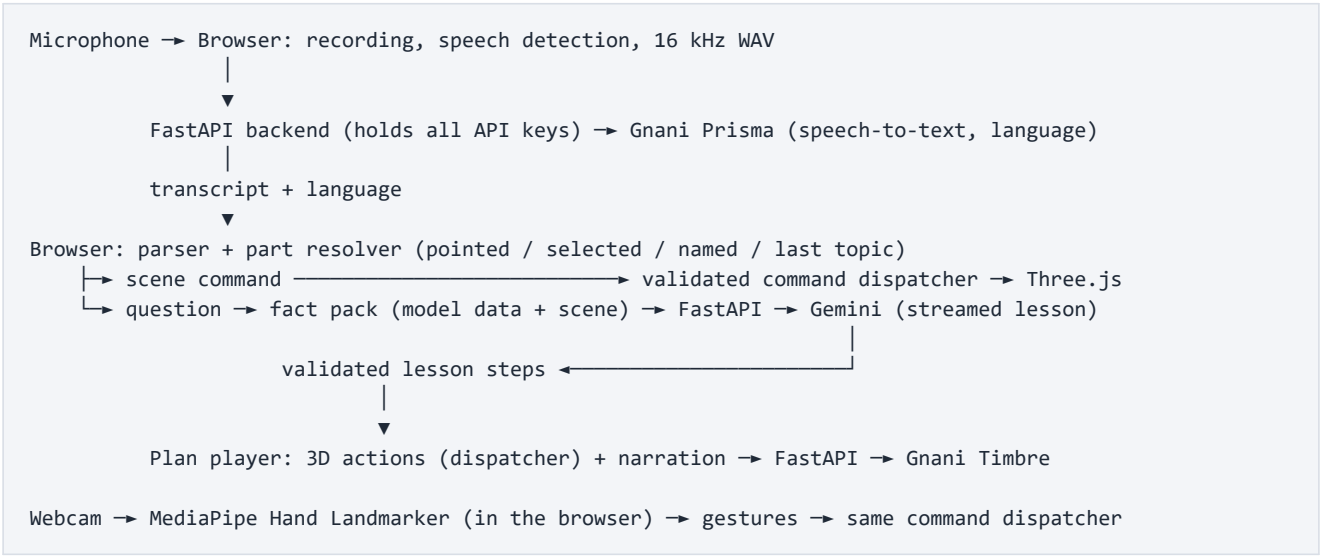


Stage	What happens	Measured in live tests
Recording	Echo cancellation and noise suppression on; a 300 ms start window ignores the tutor's own voice and the microphone's click; recording ends about 1.1 s after the learner stops. Exact digital silence is reported as a muted microphone and never sent to Prisma.	Stops automatically
Gnani Prisma	One request with automatic language detection (“indic-auto”, code-mixed format) and the part names as key terms. The transcript is shown with its language, in its own script.	0.9-3.7 s per question
Language decision	Each turn is decided afresh from Prisma's detection (the script wins if they disagree), so English → Telugu → Hindi works turn by turn.	No language menu
Answer	Simple questions are answered instantly from the model data. Explanations stream from Gemini as a validated lesson plan.	First sentence ready in about 4 s (was 15 s)
Gnani Timbre	Each sentence is synthesised ahead of playback, one request at a time, with the native voice for the language.	1.7-2.6 s per sentence
Playback	Sentences play strictly in order, never overlapping. No silence is added between them; Timbre's own short pause (about 0.3 s) gives natural pacing. Pressing the mic stops everything at once.	1-51 ms between sentences

The timings come from the project's own live tests on 6-7 November 2026; they depend on the network and on the providers' response times.

SECTION 08

Architecture, Safety and Quality



Principle	How it is achieved
One command system	Mouse, keyboard, gestures, voice and the AI teacher all send the same validated scene commands. The AI can only do what a user could do.
The AI returns data, never code	Lesson plans are JSON. Actions must be on an allowed list and name real parts; they are validated on the server and again in the browser.
Keys stay secret	The browser never calls Gnani or Gemini. Keys live only on the backend and never appear in responses, logs or error messages.
Privacy	No login, no database, no stored recordings. Camera video never leaves the device.
Honest claims	Every lesson step is labelled by its source; nothing is presented as a physics simulation.
Quality	983 frontend and 222 backend automated tests; Gnani and Gemini are simulated, so tests never spend credits.

SECTION 09

Who Benefits

Users	How GestureLab CAD helps them
Mechanical and civil engineering students	Understand a gear train, an engine, a frame or a truss by asking about the exact part they are looking at, and seeing the answer on it.
Diploma, polytechnic and ITI students	Hands-on machine learning without cut-away models or a lab, on a laptop.
Learners who think in Telugu or Hindi	Ask and hear answers in their own language, with technical terms kept in English, as engineering is taught in Indian classrooms.
Self-learners revising alone	An answer at the moment the doubt appears, not the next day.
Teachers and technical trainers	Demonstrate a machine to a whole class with a laptop and a projector, by voice or gesture.
Colleges and training centres	An interactive 3D lab for the cost of a browser, a webcam and a microphone.
Science learners and exhibitions	The human eye module, and touchless voice-and-gesture demos at science fairs.

WHY IT SCALES

A new subject is simply a new 3D model with its parts data: the tutor, voice, gestures and validation work on it unchanged. Gnani's voice models cover many Indian languages, and the app already lists ten, so adding languages after English, Telugu and Hindi is the natural next step. It runs in any modern browser, so it reaches any college with an internet connection.

SECTION 10

Current Status and Conclusion

Where the project stands today

- **Working prototype, deployed:** frontend on Vercel, backend on Render, five learning modules, voice tutor, hand gestures.
- **Proven in live tests:** real microphone capture, Gnani Prisma transcription, Gemini lesson plans and Gnani Timbre speech in English, with the first sentence ready in about 4 seconds and smooth playback.
- **Telugu and Hindi** are built into every stage and tested offline; they should be confirmed with a real voice before each demonstration.
- **Not yet done:** no classroom study and no deployment in colleges. These are the next steps, not claims.

Conclusion

GestureLab CAD starts from a simple observation: **students can see a machine, but they cannot ask it anything.** Our answer is to make the machine itself the teacher. The learner points at a part, asks in their own words and their own language, and the machine responds: the right parts light up and move while a natural voice explains, step by step, and then checks that they understood.

What makes this possible is a creative use of Gnani AI. **Gnani Prisma** does more than transcribe: it detects the learner's language and turns a spoken question into an action on a real 3D model. **Gnani Timbre** does more than read text aloud: it speaks each sentence exactly when its parts appear, in English, Telugu or Hindi.

Together with a lesson planner grounded in real model data, a strict validation layer and private, in-browser hand tracking, the result is not another chatbot, but an interactive engineering tutor that runs on hardware every student already has.

> "Every student deserves to ask *why*.
> GestureLab CAD lets them ask the machine itself."

Built with Gnani Prisma and Gnani Timbre · Great Indian AI Internship Challenge 2026 · Jugaad Genius Award entry · #GreatIndianAIInternshipChallenge #GnaniAI